A DBTL-based Bioinformatics Pipeline for Comparative Metabolic Pathway Annotation in Microalgae

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Abstract. Microalgae represent a promising yet underexplored resource for sustainable bioproduction of high-value compounds such as carotenoids and omega-3 fatty acids. Comparative metabolic pathway annotation across species is, however, hampered by data heterogeneity, incomplete genome annotations, and the absence of reproducible analytical frameworks. This paper presents the design rationale and work plan for an open-source bioinformatics pipeline that applies the Design-Build-Test-Learn (DBTL) paradigm to the comparative annotation of metabolic pathways in *Dunaliella salina* and *Nannochloropsis gaditana*. We review the state of the art, identify critical gaps in reproducibility and cross-species comparability, define four SMART objectives grounded in the project specification, and outline the proposed modular architecture aligned with FAIR principles.

Keywords: Microalgae · DBTL Cycle · Metabolic Annotation · Bioinformatics Pipeline · Reproducibility

1 Introduction and Importance of Microalgae

Microalgae are unicellular photosynthetic organisms whose metabolic versatility makes them attractive platforms for the sustainable production of high-value compounds [2]. Among the most commercially relevant products are pigments including β-carotene and astaxanthin, polyunsaturated fatty acids such as EPA and DHA, and a range of bioactive secondary metabolites, all of which can be accumulated at industrially viable concentrations under controlled growth conditions [3]. Unlike terrestrial crops, microalgae do not compete for arable land, can utilise industrial CO_2 streams, and offer generation times measured in hours, substantially accelerating the design-test cycle for metabolic engineering.

Two species stand out for their complementary biotechnological profiles. *Dunaliella salina* is a halotolerant, non-cell-walled green alga capable of accumulating carotenoids to more than 10% of its dry weight under salt stress or nitrogen limitation [2]. Its stress-response mechanisms — notably osmotic adjustment via glycerol accumulation and upregulation of antioxidant pathways — are governed by tightly regulated metabolic networks whose annotation remains

incomplete [6]. *Nannochloropsis gaditana*, by contrast, is a marine eustigmatophyte recognised for its unusually high content of eicosapentaenoic acid (EPA) and its capacity for rapid lipid accumulation, making it a priority target for algal biofuel and nutraceutical research [4]. A genome-scale metabolic model of this species has been reconstructed and used to predict lipid overproduction targets via flux balance analysis [5], yet cross-species comparisons with other microalgae remain scarce.

Despite their complementary profiles, a rigorous, quantitative comparison of the metabolic pathway repertoires of these two species is currently missing from the literature. Existing genome annotations are fragmented across heterogeneous databases, lack standardisation in nomenclature, and have been produced with different computational tools and reference sets, making cross-species inference unreliable.

Addressing this gap requires a structured, reproducible computational framework. The Design-Build-Test-Learn (DBTL) paradigm, formalised in the context of microbial biosynthesis [1], provides that structure: iterative cycles of computational design, workflow construction, empirical validation, and data-driven refinement map naturally onto the stages of a bioinformatics pipeline. This paper proposes precisely such a DBTL-grounded pipeline and articulates the design choices underlying it.

2 State of the Art

2.1 The DBTL Framework in Metabolic Engineering

The DBTL cycle was introduced as an engineering abstraction for optimising microbial biosynthetic pathways [1]. In its original formulation, the Design phase enumerates candidate pathways *in silico*; Build translates designs into biological or computational constructs; Test quantifies performance against defined criteria; and Learn applies statistical or machine-learning methods to extract actionable insights that seed the next design iteration. Carbonell *et al.* [1] demonstrated the viability of fully automated DBTL pipelines for fine chemical production, establishing a template later adopted in systems biology more broadly. Subsequent work has extended this paradigm from single-organism optimisation to comparative genomics, although applications specifically targeting microalgae remain scarce.

2.2 Bioinformatics Tools and Pipelines for Microalgae

Reproducible workflow management is a prerequisite for comparative genomic studies. Snakemake [7] has become a standard tool for this purpose, providing Python-based pipeline definition, automatic dependency resolution, and native support for containerisation via Docker or Singularity, all of which are essential for ensuring consistent execution across platforms. In the context of microalgae, Cunha *et al.* [2] surveyed systems biology approaches for microalgal biomass

and identified the lack of standardised annotation pipelines as one of the central bottlenecks to translating genomic knowledge into bioproduction strategies.

For *N. gaditana*, Shah *et al.* [5] reconstructed the first compartmentalised genome-scale metabolic model (iRJ1321), encompassing 1321 genes, 1918 reactions, and 1862 metabolites, and used it to analyse metabolic capabilities for lipid overproduction. For *D. salina*, Chen *et al.* [6] reported a high-quality genome sequence and combined genomics with transcriptomics to elucidate the β-carotene biosynthesis mechanism, performing KEGG pathway enrichment analysis that revealed key up- and down-regulated genes under high-light stress. Crucially, both studies were conducted in isolation, with no shared methodology or benchmarking framework, illustrating the absence of cross-species reproducibility in the field.

Taken together, these studies point to two recurrent challenges: (i) each pipeline is constructed *ad hoc* for a single species or compound class, and (ii) the absence of shared benchmarking datasets prevents rigorous comparison of annotation tools across species.

2.3 Key Bioinformatics Tools

Table 1 summarises the tools evaluated in this work according to the comparison matrix defined in the project objectives, covering functional scope, input/output types, open-source licence, and containerisation support.

Table 1. Comparison matrix of candidate bioinformatics tools

Tool	Phase	Function	Input	Output	Open	Container
antiSMASH	Learn	BGC annotation	Genome	Gene clusters	Yes	Yes
KEGG/ KofamScan	Design	Pathway/KO mapping	Proteins	KO terms	Partial	Yes
COBRApy	Design	Metabolic mod.	GEM	Fluxes	Yes	Yes
AlphaFold2	Learn	Structure pred.	Protein seq	3D structure	Yes	Yes
AutoDock Vina	Learn	Mol. docking	Prot.+ligand	Binding score	Yes	Yes
Clustal Omega	Learn	Seq. alignment	Sequences	MSA	Yes	Yes
Snakemake	Build	Workflow mgmt	Config	DAG execution	Yes	Yes
Biopython	All	Data I/O & parsing	Bio formats	Parsed objects	Yes	Yes

2.4 Identified Gaps

Three critical gaps emerge from the review. First, **data heterogeneity**: genomic, transcriptomic, and metabolomic datasets for microalgae are distributed across incompatible databases (NCBI, JGI, KEGG, UniProt) with inconsistent gene identifiers and ontology mappings. Second, **lack of comparative reproducibility**: no published pipeline provides a standardised, end-to-end workflow for cross-species metabolic pathway comparison in microalgae. Third, **insufficient benchmarking**: existing tools are typically evaluated on model organisms (*E. coli*, *S. cerevisiae*) and their performance on microalgal genomes — characterised by high proportions of hypothetical proteins and incomplete assemblies — is rarely reported. These gaps collectively motivate the present work.

3 Problem Statement and Objectives

The central problem addressed by this project can be stated precisely: *incomplete and heterogeneous metabolic pathway annotations prevent reliable cross-species comparisons in microalgae, impeding the identification of species-specific biosynthetic capabilities relevant to bioproduction.*

The absence of a reproducible, standardised pipeline means that every research group must rebuild the analytical infrastructure from scratch, introducing variability that is methodological in origin rather than biological. This problem is compounded by the partial open-source coverage of key databases (notably KEGG) and the rapid evolution of annotation tools, which makes version-locked analyses difficult to reproduce.

To address this, the project defines four SMART objectives:

1. **Survey and critically evaluate** at least six existing tools for genome annotation, pathway prediction, and metabolomics integration, building a structured comparison matrix covering scope, inputs, outputs, open-source licences, API availability, containerisation, and interoperability.
2. **Select and integrate** two complementary tools — a genome annotation tool and a KEGG orthology mapping tool for pathway prediction — into a reproducible Snakemake workflow, containerised with Docker or Singularity.
3. **Annotate and compare** the metabolic pathway repertoires of *D. salina* and *N. gaditana*, focusing on carotenoid and omega-3 fatty acid biosynthesis pathways, depositing all outputs in a public repository.
4. **Benchmark** pipeline accuracy using sensitivity, specificity, and F1 score against curated annotations from UniProt/Swiss-Prot and published species-specific literature.

4 Proposed Approach

4.1 DBTL Framework Mapping

The pipeline architecture maps onto the four DBTL phases as follows.

Design. Candidate biosynthetic pathways are enumerated *in silico* using KEGG pathway maps and KofamScan for KO-based orthology assignment. COBRApy is used to construct and interrogate genome-scale metabolic models (GEMs), enabling flux balance analysis (FBA) to predict flux distributions under relevant growth conditions. Tool selection follows the comparison matrix in Table 1.

Build. Tools are integrated into a Snakemake directed acyclic graph (DAG) [7]. Docker containers encapsulate each tool with its exact dependencies, ensuring reproducible execution across platforms. RO-Crate packages the workflow with EDAM ontology annotations to ensure semantic interoperability [11]. All pipeline versions, input data, results, and supplementary material are deposited in a public GitHub repository.

Test. Predicted pathway annotations are evaluated against curated databases and manually verified literature entries. The primary metrics — sensitivity, specificity, and F1 score — are computed against a holdout gold-standard set of biochemically characterised enzymes from both species drawn from UniProt/Swiss-Prot.

Learn. Discrepancies between predicted and curated annotations are systematically catalogued and fed back into the Design phase. Hypothetical proteins with strong structural or docking evidence (AlphaFold2 predictions scored via AutoDock Vina) are flagged for re-annotation, and the cycle is repeated until performance plateaus.

4.2 Modularity and FAIR Compliance

The modular structure — one Snakemake rule per analytical step — allows individual modules to be replaced or extended without disrupting the remainder of the workflow. FAIR compliance [11] is ensured through persistent identifiers for all input datasets, open licensing (MIT), versioned container images, and RO-Crate provenance records. This design directly targets the reproducibility and heterogeneity gaps identified in Section 2.

5 Work Plan

The project is structured into four sequential phases, each producing a concrete deliverable that feeds into the next and enables preliminary validation before full pipeline deployment (Table 2).

Table 2. Project phases, activities, and deliverables

Ph.	Phase name	Activities	Deliverable
1	Literature & Tool Survey	Systematic literature review; tool evaluation; comparison matrix; pipeline architecture	Intercalar article; GitHub repository initialised
2a	Build	Snakemake DAG; Docker containers; EDAM annotation; RO-Crate packaging	Functional containerised pipeline v0.1
2b	Test & Preliminary Results	Run pipeline on both species; compute sens./spec./F1; prepare presentation slides	Public presentation; preliminary benchmarking results
3	Analysis & Final Report	Cross-species pathway analysis; gap identification; iterative refinement; documentation	Final article; complete public repository

Phases 2a and 2b are internally sequential; analysis begun in 2b is completed and extended in Phase 3, allowing presentation feedback to be incorporated into the final article. Each transition point includes an internal review against the DBTL Learn phase, ensuring that preliminary results inform subsequent design iterations.

6 Conclusion

This paper has argued that accurate comparative annotation of metabolic pathways in microalgae requires a structured, reproducible pipeline grounded in a principled iterative methodology. The DBTL framework provides that structure, mapping cleanly onto genome annotation, workflow construction, empirical validation, and data-driven refinement. The proposed pipeline addresses three concrete gaps identified in the literature — data heterogeneity, absence of cross-species reproducibility, and insufficient benchmarking on microalgal genomes. By targeting *Dunaliella salina* and *Nannochloropsis gaditana*, species with complementary and commercially relevant biosynthetic profiles in carotenoids and omega-3 fatty acids, the work connects fundamental annotation challenges directly to cell engineering and bioproduction priorities. The four SMART objectives and phased work plan ensure that each intermediate deliverable is independently verifiable within the scope of this project.

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